

Environmental exposure to pesticides and cancer risk in multiple human organ systems.

There is growing evidence on the association between long-term exposure to pesticides in occupational settings and an elevated rate of chronic diseases, including different types of cancer. However, data on non-occupational exposures are scarce to draw any conclusion. The objective of this study was to investigate the putative associations of environmental pesticide exposures in the general population with several cancer sites and to discuss potential carcinogenic mechanisms by which pesticides develop cancer. A population-based case-control study was conducted among people residing in 10 Health districts from Andalusia (South Spain) to estimate the risk of cancer at different sites. Health districts were categorized into areas of high and low environmental pesticide exposure based on two quantitative criteria: number of hectares devoted to intensive agriculture and pesticide sales per capita. The study population consisted of 34,205 cancer cases and 1,832,969 age and health district matched controls. Data were collected by computerized hospital records (minimum dataset) between 1998 and 2005. Prevalence rates and the risk of cancer at most organ sites were significantly higher in districts with greater pesticide use related to those with lower pesticide use. Conditional logistic regression analyses showed that the population living in areas with high pesticide use had an increased risk of cancer at all sites studied (odds ratios between 1.15 and 3.45) with the exception of Hodgkin's disease and non-Hodgkin lymphoma. The results of this study support and extend previous evidence from occupational studies indicating that environmental exposure to pesticides may be a risk factor for different types of cancer at the level of the general population. Copyright © 2013 Elsevier Ireland Ltd. All rights reserved.